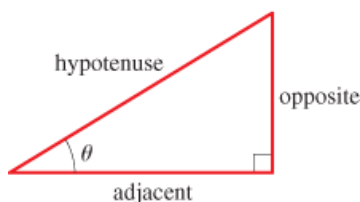


I. Trigonometric Ratios

Consider a right triangle with θ as one of its acute angles. Complete the following trigonometric ratios.



SOH/CAH/TOA

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{\csc \theta}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{1}{\sec \theta}$$

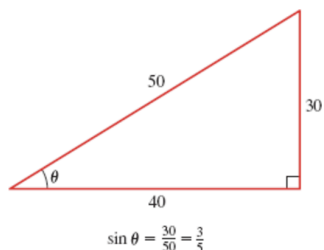
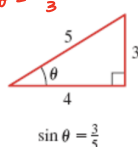
$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin \theta}{\cos \theta}$$

$$\csc \theta = \frac{\text{hypotenuse}}{\text{opposite}} = \frac{1}{\sin \theta}$$

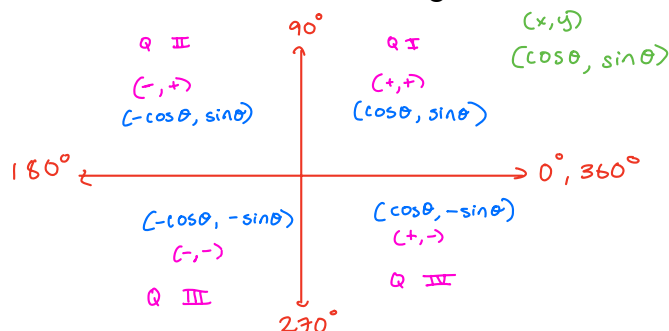
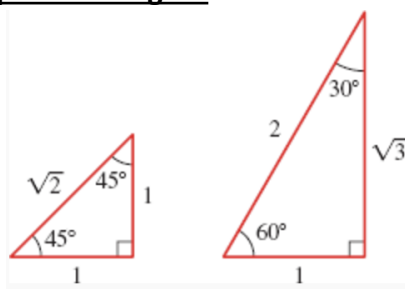
$$\sec \theta = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{\text{adjacent}}{\text{opposite}} = \frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta}$$

$$\begin{aligned} \cos \theta &= \frac{4}{5}, \sec \theta = \frac{5}{4} \\ \tan \theta &= \frac{3}{4}, \cot \theta = \frac{4}{3} \end{aligned}$$

**II. Special Triangles ; Calculators**

Certain right triangles have ratios that can be calculated from certain triangles.

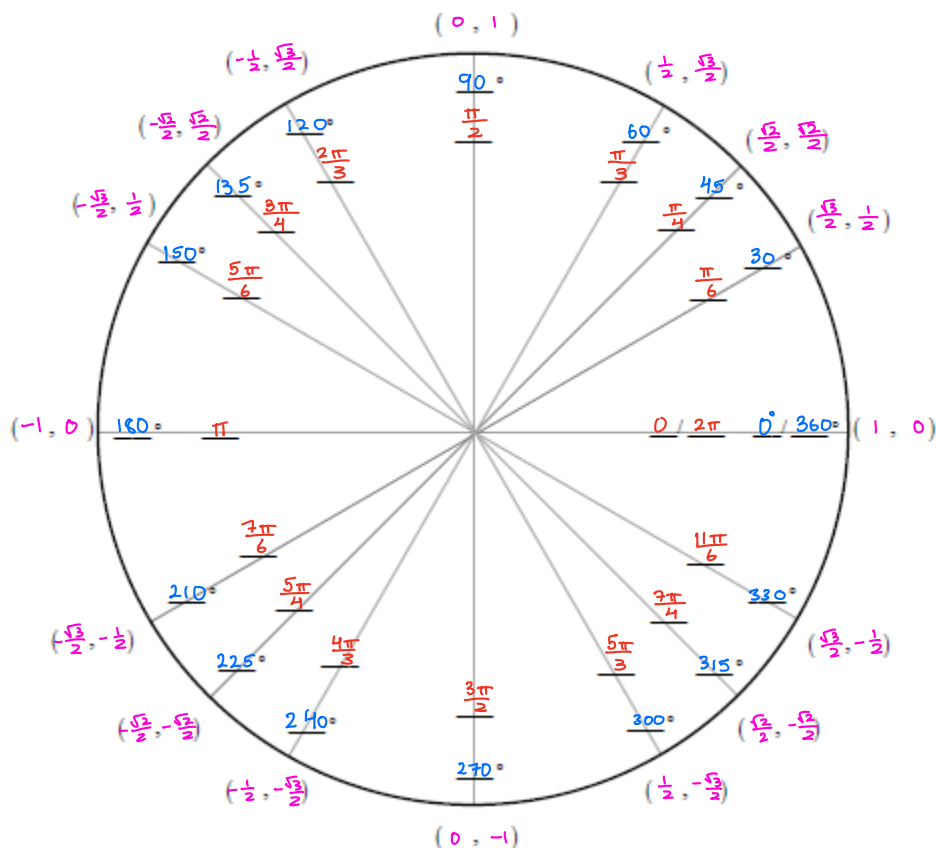
Special Triangles:

Quadrantal Numbers and Angles: $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ or $\theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$

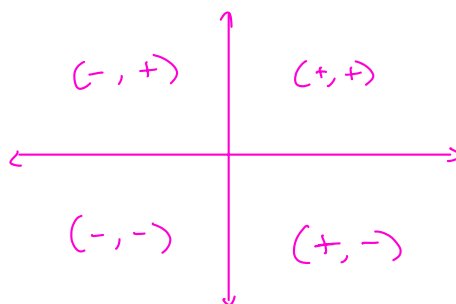
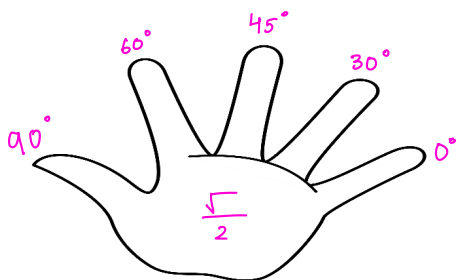
Special Values of the Trigonometric Functions

Complete the following table of values of the trigonometric ratios for special angles.

θ in degrees	θ in radians	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\csc \theta$	$\sec \theta$	$\cot \theta$
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	2	$\frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$	$\sqrt{3}$
45°	$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$	2	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$



Here is a way to remember your unit circle which we called the Left-Hand Trick

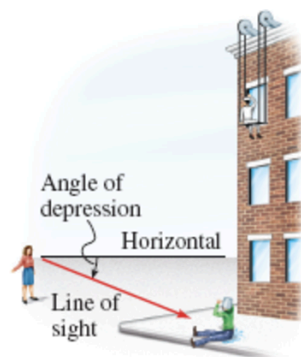


Calculators give the values of sine, cosine, and tangent; the other ratios can be easily calculated from these by using reciprocal relationships.

$$\csc t = \frac{1}{\sin t}, \sec t = \frac{1}{\cos t}, \cot t = \frac{1}{\tan t}$$

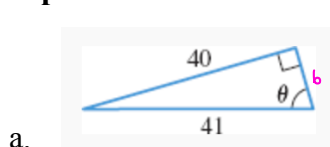
III. Applications of Trigonometry of Right Triangles

A triangle has six parts: three angles and three sides. To solve a triangle means to determine all of its parts from the information known about the triangle, that is, to determine the lengths of the three sides and the measures of the three angles.



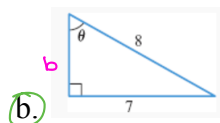
If an observer is looking at an object, then the line from the eye of the observer to the object is called the line of sight. If the object being observed is above the horizontal, then the angle between the line of sight and the horizontal is called the angle of elevation. If the object is below the horizontal, then the angle between the line of sight and the horizontal is called the angle of depression. If the line of sight follows a physical object, such as an inclined plane or hillside, we use the term angle of inclination.

Example 1: Find the exact values of the six trigonometric ratios of the angle θ in the triangle.



$$b = \sqrt{41^2 - 40^2} = 9$$

$$\begin{aligned} \sin \theta &= \frac{40}{41}, & \csc \theta &= \frac{41}{40} \\ \cos \theta &= \frac{9}{41}, & \sec \theta &= \frac{41}{9} \\ \tan \theta &= \frac{40}{9}, & \cot \theta &= \frac{9}{40} \end{aligned}$$

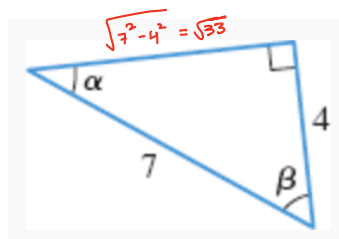


$$b = \sqrt{8^2 - 7^2} = \sqrt{15}$$

$$\begin{aligned} \sin \theta &= \frac{7}{8}, & \csc \theta &= \frac{8}{7} \\ \cos \theta &= \frac{\sqrt{15}}{8}, & \sec \theta &= \frac{8}{\sqrt{15}} = \frac{8\sqrt{15}}{15} \\ \tan \theta &= \frac{7}{\sqrt{15}} = \frac{7\sqrt{15}}{15}, & \cot \theta &= \frac{\sqrt{15}}{7} \end{aligned}$$

Example 2: Find

a. $\sin \alpha$ and $\cos \beta$



$$\sin \alpha = \frac{4}{7}$$

$$\cos \beta = \frac{4}{7}$$

b. $\tan \alpha$ and $\cot \beta$

$$\begin{aligned} \tan \alpha &= \frac{4}{\sqrt{33}} = \frac{4\sqrt{33}}{33} \\ \cot \beta &= \frac{4}{\sqrt{33}} = \frac{4\sqrt{33}}{33} \end{aligned}$$

c. $\sec \alpha$ and $\csc \beta$

$$\begin{aligned} \sec \alpha &= \frac{7}{\sqrt{33}} = \frac{7\sqrt{33}}{33} \\ \csc \beta &= \frac{7}{\sqrt{33}} = \frac{7\sqrt{33}}{33} \end{aligned}$$

Example 3: Use a calculator in degree or radian mode, as appropriate, to evaluate each expression. Round your answer to five decimal places.

a. $\sin 3.5$

$= -0.35078$

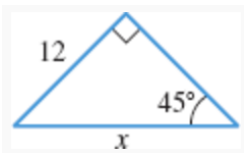
b. $\csc 48^\circ$

$= \frac{1}{\sin 48^\circ} = 1.34563$

c. $\sec 1$

$= \frac{1}{\cos 1} = 1.85082$

Example 4: Find the side labeled x .



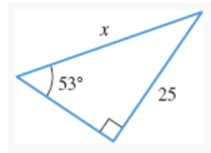
a.

$$\sin 45^\circ = \frac{12}{x}$$

$$x = \frac{12}{\sin 45^\circ}$$

$$x = \frac{12}{\frac{\sqrt{2}}{2}}$$

$$x = 12\sqrt{2} \text{ units}$$



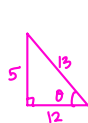
b.

$$\sin 53^\circ = \frac{25}{x}$$

$$x = \frac{25}{\sin 53^\circ} \text{ units}$$

Example 5: Sketch a triangle that has acute angle θ , and find the other five trigonometric ratios of θ .

a. $\cos \theta = \frac{12}{13}$



$\tan \theta = \frac{5}{12}$

$\sin \theta = \frac{5}{13}$

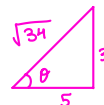
$\csc \theta = \frac{13}{5}$

$\sec \theta = \frac{13}{12}$

$\cot \theta = \frac{12}{5}$

b. $\csc \theta = \frac{11}{6}$

c. $\cot \theta = \frac{5}{3}$, $\tan \theta = \frac{3}{5}$



$\sin \theta = \frac{3}{\sqrt{34}} = \frac{3\sqrt{34}}{34}$

$\csc \theta = \frac{\sqrt{34}}{3}$

$\cos \theta = \frac{5}{\sqrt{34}} = \frac{5\sqrt{34}}{34}$

$\sec \theta = \frac{\sqrt{34}}{5}$

Example 6: Evaluate the expression without using a calculator.

a. $\sin 30^\circ \cos 60^\circ + \sin 60^\circ \cos 30^\circ$

$$\frac{1}{2} \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{1}{4} + \frac{3}{4}$$

$$= 1$$

b. $(\cos 30^\circ)^2 - (\sin 30^\circ)^2$

$$\left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{1}{2}\right)^2$$

$$= \frac{3}{4} - \frac{1}{4}$$

$$= \frac{1}{2}$$

b. $\left(\sin \frac{\pi}{3} \tan \frac{\pi}{6} + \csc \frac{\pi}{4}\right)^2$

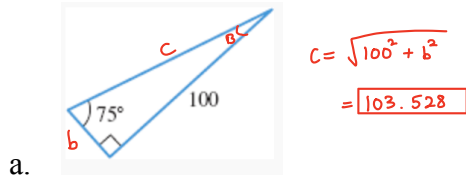
$$\left(\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}} + \sqrt{2}\right)^2$$

$$= \left(\frac{1}{2} + \sqrt{2}\right)^2 \rightarrow \text{FOIL}$$

$$= \frac{1}{4} + \sqrt{2} + 2$$

$$= \frac{9}{4} + \sqrt{2}$$

Example 7: Solve the right triangle.

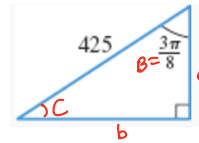


a.

$$\begin{aligned} m\angle B &= 90 - 75^\circ = 15^\circ \\ \tan 15^\circ &= \frac{b}{100} \\ 100 \cdot \tan 15^\circ &= b \\ \therefore b &= 26.795 \end{aligned}$$

$$\begin{aligned} c &= \sqrt{100^2 + b^2} \\ &= 103.528 \end{aligned}$$

b.



$$\begin{aligned} m\angle C &= \frac{\pi}{2} - \frac{3\pi}{8} = \frac{\pi}{8} \rightarrow \frac{\pi}{8} \cdot \frac{180^\circ}{\pi} = 22.5^\circ \\ m\angle B &= \frac{3\pi}{8} \cdot \frac{180^\circ}{\pi} = 67.5^\circ \end{aligned}$$

$$\begin{aligned} \cos(67.5^\circ) &= \frac{c}{425} \\ c &= 425 \cdot \cos(67.5^\circ) \end{aligned}$$

$$c \approx 162.640 \text{ m}$$

$$\begin{aligned} \cos(22.5^\circ) &= \frac{b}{425} \\ b &= 425 \cdot \cos(22.5^\circ) \\ b &\approx 392.649 \text{ m} \end{aligned}$$

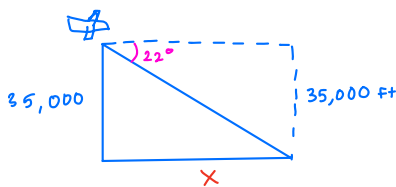
Example 8: The angle of elevation to the top of the Empire State Building in New York is found to be 11° from the ground at a distance of one mile from the base of the building. Using this information, find the height of the Empire State Building.



$$\begin{aligned} \tan 11^\circ &= \frac{h}{1 \text{ mi}} \\ h &= \tan 11^\circ \cdot 5280 \text{ ft} \\ &= 1026.328 \text{ ft} \end{aligned}$$

Example 9: A plane is flying within sight of an airport, at an elevation of 35,000 ft. The pilot would like to estimate the plane's distance to a point on the runway and finds that the angle of depression to that point is 22° .

(a) What is the distance between the plane and the point on the runway?



$$\csc 22^\circ = \frac{c}{35,000}$$

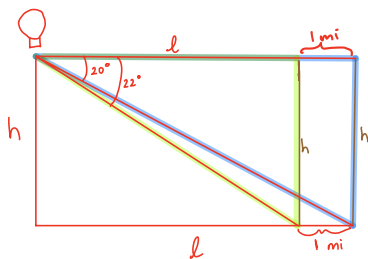
$$c = 35,000 \frac{\csc 22^\circ}{\sin 22^\circ}$$

$$c = 93431.351 \text{ ft}$$

(b) What is the distance between a point on the ground directly below the plane and the point on the runway?

$$\begin{aligned} x &= \sqrt{c^2 - 35,000^2} \\ &= 86628.04 \text{ ft} \end{aligned}$$

Example 10: A hot-air balloon is floating above a straight road. To estimate their height above the ground, the balloonists simultaneously measure the angle of depression to two consecutive mileposts on the road on the same side of the balloon. The angles of depression are found to be 20° and 22° . How high is the balloon?



Triangle 1 ■

$$\tan 20^\circ = \frac{h}{l+1} \rightarrow l+1 = \frac{h}{\tan 20^\circ}$$

Triangle 2 ■

$$\tan 22^\circ = \frac{h}{l} \rightarrow l = \frac{h}{\tan 22^\circ}$$

$$\frac{h}{\tan 22^\circ} + 1 = \frac{h}{\tan 20^\circ}$$

$$1 = \frac{h}{\tan 20^\circ} - \frac{h}{\tan 22^\circ}$$

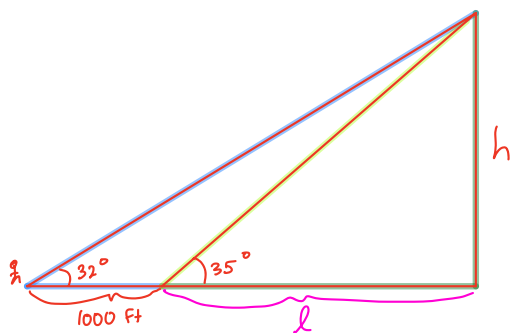
$$1 = h \left(\frac{1}{\tan 20^\circ} - \frac{1}{\tan 22^\circ} \right)$$

$$h = \frac{1}{\frac{1}{\tan 20^\circ} - \frac{1}{\tan 22^\circ}}$$

$$h \approx 3.671 \text{ mi}$$

The balloon is
 ≈ 3.671 mi high.

Example 11: To estimate the height of a mountain above a level plain, the angle of elevation to the top of the mountain a surveyor measures to be 32° . One thousand feet closer to the mountain along the plain, the angle of elevation is found to be 35° . Estimate the height of the mountain.



Triangle 1 ■

$$\tan 32^\circ = \frac{h}{l+1000} \rightarrow l+1000 = \frac{h}{\tan 32^\circ}$$

Triangle 2 ■

$$\tan 35^\circ = \frac{h}{l} \rightarrow l = \frac{h}{\tan 35^\circ}$$

$$\frac{h}{\tan 35^\circ} + 1,000 = \frac{h}{\tan 32^\circ}$$

$$1000 = \frac{h}{\tan 32^\circ} - \frac{h}{\tan 35^\circ}$$

$$1000 = h \left(\frac{1}{\tan 32^\circ} - \frac{1}{\tan 35^\circ} \right)$$

$$h = \frac{1000}{\frac{1}{\tan 32^\circ} - \frac{1}{\tan 35^\circ}}$$

$$h \approx 5807.655 \text{ feet}$$

The height of the
mountain is ≈ 5807.655 ft.